

Operating System Lab Assignment – 3



Comprehensive Implementation of Page Replacement Algorithms

Submitted By:

Tarun
BCA (AI & Data Science) - B
2401201077

1. Introduction

In modern operating systems, virtual memory plays a crucial role in managing system resources efficiently. When the number of pages exceeds the available physical memory frames, page replacement algorithms are used to decide which page should be removed from memory.

These algorithms aim to minimize page faults and improve system performance. This assignment focuses on implementing and analyzing different page replacement algorithms such as FIFO, LRU, Optimal, MRU, and Second Chance.

2. Objectives

- To understand the concept of paging and virtual memory
 - To implement different page replacement algorithms
 - To calculate and compare page faults
 - To analyze performance of each algorithm
-

3. Tools & Technologies Used

- Python 3.x
 - VS Code / PyCharm
 - Basic Python Libraries
 - Ubuntu/Linux Terminal
-

4. Input Details

- Number of Frames: (example: 3)
 - Page Reference String: (example: 7 0 1 2 0 3 0 4 2 3 0 3 2)
-

5. Algorithms Implemented

5.1 FIFO (First-In First-Out)

FIFO replaces the oldest page present in memory. It uses a queue structure where the first page inserted is the first to be removed.

Advantages:

- Simple to implement

Disadvantages:

- May lead to more page faults
-

5.2 LRU (Least Recently Used)

LRU replaces the page that has not been used for the longest time.

Advantages:

- Uses real-life usage pattern
- More efficient than FIFO

Disadvantages:

- Requires tracking of page usage
-

5.3 Optimal Page Replacement

This algorithm replaces the page that will not be used for the longest time in the future.

Advantages:

- Produces minimum page faults

Disadvantages:

- Not practical (future knowledge required)

5.4 MRU (Most Recently Used)

MRU replaces the most recently used page.

Advantages:

- Useful in specific scenarios

Disadvantages:

- Not commonly used
-

5.5 Second Chance Algorithm

This is an improved version of FIFO. It gives a second chance to pages before replacing them.

Advantages:

- Better than FIFO
- Reduces unnecessary replacements

Disadvantages:

- Slightly complex
-

6. Output Results

Algorithm	Page Faults
FIFO	XX
LRU	XX
Optimal	XX
MRU	XX
Second Chance	XX

(Replace XX with your actual output values)

7. Performance Comparison

- Optimal algorithm gives the least number of page faults
 - LRU performs better than FIFO in most cases
 - MRU performs poorly in general scenarios
 - Second Chance improves FIFO performance
-

8. Result Analysis

From the results, it is observed that the Optimal algorithm performs best because it uses future knowledge to make decisions. However, it is not feasible in real systems.

LRU is the most practical algorithm as it approximates real usage behavior. FIFO is simple but less efficient, while Second Chance provides a better alternative to FIFO.

9. Conclusion

This assignment helped in understanding the working of various page replacement algorithms and their impact on system performance. It also demonstrated how different strategies affect the number of page faults.

Among all, LRU is widely used in real systems due to its balance between performance and practicality.

10. References

- Operating System Concepts by Silberschatz
 - Lecture Notes
 - Python Documentation
-